Artem Kolos

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Kaggle: kaggle.com/artemkolos/code



Profile

Backend Engineer with nearly 5 years of experience designing data pipelines, worker queues, and APIs in Python and Go. Recently built AI-powered scraping and labeling systems for researchers and the backend of a large-scale AI support platform. Passionate about distributed systems.

Work Experience

Software Engineer
Academia Sinica, Communication Data and Network Analytics Lab (CDNA)
06/2026 – Present Taipei, Taiwan

- Develop CLI tools and APIs in Go that let data scientists scrape, label, and export research data.
- Built Go worker queues backed by PostgreSQL and SQLite for concurrent, resumable scraping and labeling jobs.
- Set up tracing with OpenTelemetry and Grafana to monitor LLM token consumption, pipeline throughput, and latency.

Technologies: Go, PostgreSQL, SQLite, OpenTelemetry, Grafana, GCP
Personal Achievements: Built a pipeline that scrapes, labels, and then summarizes data with AI, processing 20,000–30,000 rows of complex, context-dependent text per day across thousands of concurrent jobs.

Backend Engineer
Seasalt.ai
04/2025 – 07/2026 Taipei, Taiwan

- Designed, maintained, and optimized a microservice-based backend architecture powering a large-scale AI customer support platform (chat + voice).
- Managed and extended integrations with third-party CRMs and major communication channels, including WhatsApp, Facebook, Instagram, LINE, Twilio, and SMS providers.

Technologies: FastAPI, SQLAlchemy, PostgreSQL, Elasticsearch, Redis Queue, Prometheus, React Native, Kubernetes, Docker, AWS, Twilio
Personal Achievements: Developed and deployed a unified notification system with full support for Apple Push Notifications, Android FCM, and browser push via the Web Push API, handling over 200,000 messages per month.

Software Engineer
Processica
07/2023 – 04/2025 Remote, United States

- Designed a RAG-powered database architecture for image retrieval and dynamic prompt adaptation in a storytelling platform.
- Implemented a custom payment system for monetizing an AI chat app with one-time payments and subscriptions.
- Built a real-time API for detecting advanced stock trading patterns, using a scalable Pub/Sub architecture and integrating Databento for live market data.

Technologies: FastAPI, PostgreSQL, Pgvector, SQLAlchemy, Redis Queue, reactJS, Kubernetes, Docker, Azure DevOps, Stripe, Databento
Personal Achievements: Created a dynamic prompting workflow using PostgreSQL with PgVector to model graph structures, enabling real-time prompt updates via WebSocket based on chat history.

Work Experience

Python Developer / Data Analyst
Post Data
02/2022 – 07/2023 Remote, Australia

- Conducted web scraping and statistical analysis projects.
- Developed data-driven web apps with interactive dashboards.
- Integrated AI chatbots with custom RAG systems.
- Performed monthly time series analysis on retail data, generating interactive reports.

Technologies: Pandas, R, BeautifulSoup, Selenium, Tableau, PowerBI, Streamlit, Django, Langchain, HuggingFace, OpenAI
Personal Achievements: Developed a backend application that automates the generation of customizable visualizations and data reports, enabling flexible filtering and data retrieval for non-technical users.

Open Source

- **Starlette:** reporting bugs and improving the SSE and background tasks implementation, focusing on client disconnects in streaming responses.
- **youtube-scraper:** concurrent Go CLI that scrapes YouTube videos and comments via the Data API v3 into SQLite, with resumable jobs and convenient exports for analysis in DuckDB and similar tools. Actively used by data analysts.
- **polybar-world-clock:** customizable multi-time-zone clock widget for Polybar (a status bar for Linux desktops).

Skills

- **Languages:** Python, Go, C, JavaScript
- **Database Management:** PostgreSQL, Redis, MongoDB, Neo4j, CockroachDB
- **Messaging:** Kafka, Redis Streams
- **DevOps:** Kubernetes, Docker, Azure, AWS, GCP, Hetzner
- **Operating Systems:** Linux (Arch, Debian)
- **Generative AI:** Langchain, HuggingFace, OpenAI
- **Languages:** English (TOEFL score 105), Chinese (HSK level 6)

Education

Electronic Engineering | BSc
National Technical University of Ukraine "Igor Sikorsky Kyiv Polytechnic Institute"
09/2012 – 06/2016 Ukraine